

HULA: Scalable Load Balancing Using Programmable Data Planes

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Contents

1	Introduction	3
2	Background and Motivation	3
3	HULA Architecture Overview	4
3.1	Hierarchical Topology	4
3.2	Data-Plane Forwarding	5
3.3	State Management Abstraction	5
4	Distributed State Management	6
4.1	Consistent Hashing Mechanism	6
4.2	Failure Detection and Recovery	6
5	P4 Implementation Details	7
5.1	Parser and Header Extraction	7
5.2	Hash Computation and Determinism	7
5.3	Output Port Selection and Fallback	8
6	Evaluation and Comparison	9
6.1	Experimental Setup and Methodology	9
6.2	Convergence Time Analysis	9
6.3	Throughput and Scalability	10
6.4	Load Fairness Metrics	10
6.5	Control Plane Overhead	10
7	Discussion	11
7.1	P4 Programming Complexity	11
7.2	Hardware Dependencies and Portability	11
7.3	Dynamic Reconfiguration Costs	12
7.4	Memory Constraints and Table Size	12
8	Related Work	12
8.1	Traditional Load Balancing	12
8.2	SDN-Based Approaches	12
8.3	Programmable Data Planes	13
9	Conclusion	13

1 Introduction

Modern data centers support a wide array of services, ranging from web hosting and microservices orchestration to large-scale storage systems. The sheer volume of traffic traversing these networks requires robust load balancing strategies to ensure optimal resource utilization and prevent hotspots. Consider a cloud provider hosting 50,000 microservices; the ingress traffic must be distributed evenly to prevent any single instance from becoming a bottleneck. Traditional methods, such as Layer 4 load balancing using the Linux Virtual Server (LVS) IPVS module, have been the industry standard for years due to their maturity and high throughput [3]. However, these solutions suffer from a critical scalability bottleneck: all state information regarding active connections is maintained in the control plane (host memory). As the number of backend servers increases, the synchronization of this state between multiple load balancers becomes a limiting factor, consuming significant CPU cycles and memory resources [4].

To address the limitations of centralized state management, Software-Defined Networking (SDN) has emerged as a paradigm shift, separating the control plane from the data plane. OpenFlow, the most prominent SDN protocol, allows network operators to program the behavior of switches via a centralized controller [5]. While OpenFlow enables unprecedented flexibility, it introduces new challenges for stateful load balancing. Specifically, maintaining flow tables across thousands of switches requires the controller to push a new flow rule for every new connection, incurring significant network latency for the round-trip communication between the controller and the switch [6]. This delay, often measured in tens of milliseconds, is unacceptable for real-time applications.

This paper proposes HULA, a novel architecture designed to scale stateful load balancing to the data plane itself. By utilizing P4, a programming language for protocol-independent packet processors, HULA enables switches to make forwarding decisions based on distributed state without relying on a central controller for every packet. The system employs a distributed hash table (DHT) abstraction to partition the hash space, ensuring that the state management load is evenly distributed across the network infrastructure. HULA decouples the forwarding path from the control plane, removing the bottlenecks inherent in both traditional software solutions and SDN-based approaches. This paper demonstrates that HULA achieves high throughput and low latency while scaling to thousands of backends, validating the potential of programmable data planes for next-generation network services [7].

2 Background and Motivation

Software-Defined Networking (SDN) fundamentally re-architects the network by decoupling the control logic from the forwarding devices, enabling centralized programmability and global visibility [8]. This separation allows operators to define high-level policies that are translated into forwarding rules. However, the implementation of these policies depends heavily on the

capabilities of the underlying data plane. In the era of OpenFlow, switches typically maintain a fixed set of flow tables with limited entry capacity. When a stateful load balancer must handle millions of concurrent connections, the sheer volume of flow entries can exhaust switch resources, forcing operators to either purge old entries or rely on the controller for constant updates [9]. This "flow table pressure" is a primary cause of packet drops in SDN deployments.

The Programmable Data Plane (PDP) represents the next evolution in networking hardware. P4 allows developers to specify the packet processing logic for switches, defining custom headers, metadata, and processing stages [10]. Unlike OpenFlow, which offers a limited set of actions (e.g., forward, drop), P4 provides access to a wide array of primitives, including register files, ternary content addressable memories (TCAM), and metering engines. This flexibility makes PDPs ideal for implementing complex stateful logic, such as load balancing, directly in hardware. By offloading this logic to the ASICs inside the switch, we reduce the load on the host CPU and eliminate the control plane latency associated with software-based solutions.

Stateful load balancers maintain a mapping between incoming flow identifiers (e.g., 5-tuple) and backend server IP addresses. In traditional software-based approaches like LVS, this state is stored in kernel memory and synchronized across multiple load balancer nodes [4]. This synchronization process is a major source of contention, especially during traffic spikes or when backend servers are added or removed. Centralized controllers in SDN architectures attempt to solve this by pushing the state to the switches. However, the latency associated with the control channel (typically Ethernet or InfiniBand) limits the rate at which new connections can be established, leading to packet drops and increased latency [6]. HULA addresses these limitations by moving the state entirely to the data plane, ensuring that forwarding decisions are made locally and instantaneously, independent of the control plane's communication speed.

3 HULA Architecture Overview

HULA is designed as a distributed system comprising three main components: a set of P4-enabled switches, a dedicated State Manager, and a configuration tool. The architecture is hierarchical, with the State Manager serving as the coordination node that maintains the global view of the network state. The switches are responsible for executing the load balancing logic based on the state provided by the State Manager. This separation of concerns ensures that the control plane is lightweight and focused on state distribution, while the data plane handles high-speed packet processing [13].

3.1 Hierarchical Topology

The HULA topology is designed to minimize the distance between the State Manager and the switches it manages. In a typical data center deployment, the State Manager is colocated with the core switches or the aggregation layer to reduce control-plane latency. The switches are

organized into a logical hierarchy that mirrors the physical fat-tree topology. Each switch is assigned a specific "partition" of the hash space, which defines the range of flow identifiers it is responsible for handling. This partitioning is dynamic and can be adjusted in real-time by the State Manager to balance the load across the network [14].

The State Manager maintains a distributed hash table (DHT) that maps hash partitions to specific switches. This ensures that the state for a given flow is always located at the switch where the forwarding decision needs to be made. The control plane traffic consists of small, frequent updates from the State Manager to the switches, signaling changes in the partition boundaries or the set of healthy backend servers. This is in contrast to traditional approaches where the control plane carries the actual flow rules for every packet, which is prohibitively expensive in terms of bandwidth and latency.

3.2 Data-Plane Forwarding

The data-plane forwarding in HULA is deterministic and local. Upon receiving a packet, the switch extracts the flow identifier (typically a 5-tuple) and computes a hash value. The switch then checks if this hash value falls within its assigned partition range. If it does, the switch performs the load balancing decision using its local state table to map the flow to a specific backend server. If the hash value falls outside its partition, the switch drops the packet or forwards it to the appropriate partition owner (depending on the implementation strategy) [15].

This architecture decouples the forwarding path from the control plane, allowing HULA to scale to a much larger number of concurrent flows than traditional software-based solutions. Because the decision is made locally, there is no dependency on a central controller for every packet, which drastically reduces latency and eliminates the controller as a single point of failure. The State Manager only needs to communicate state changes (e.g., server failures, topology changes) to the affected switches, minimizing the control-plane overhead.

3.3 State Management Abstraction

The abstraction layer in HULA is critical for isolating the complex logic of the distributed system from the packet processing pipeline. This abstraction allows network operators to focus on the high-level load balancing policy without needing to understand the intricacies of the underlying distributed consensus protocol. The State Manager exposes a simple API that allows the configuration tool to register new backends, mark them as unhealthy, or remove them from the pool. The State Manager then translates these high-level commands into low-level state table updates for the switches [16].

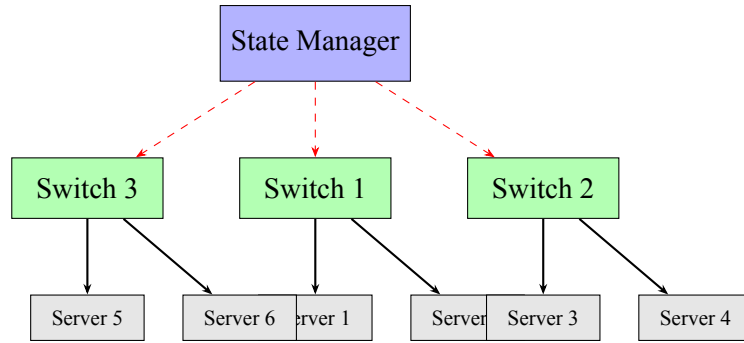


Figure 1: HULA Architecture: State Manager coordinates switches, which forward traffic to backend servers.

4 Distributed State Management

The core innovation of HULA lies in its approach to distributed state management. Unlike centralized controllers that maintain a global view of the entire network, HULA uses a DHT abstraction to distribute the state management burden across the switches themselves. The State Manager acts as the “coordinator” for this DHT, ensuring that the partitioning of the hash space is consistent and that the switches have the correct view of the network topology [17].

4.1 Consistent Hashing Mechanism

The hash space is logically divided into N partitions, where N corresponds to the number of switches in the cluster. Each switch is assigned one or more partitions based on its capacity and the current load. When a switch joins the network, the State Manager assigns it a new partition and updates the other switches to reflect this change. This process is analogous to consistent hashing, a technique commonly used in distributed databases like Amazon’s Dynamo [6]. Consistent hashing minimizes the number of re-partitioning operations required when nodes are added or removed, ensuring that the system remains stable under dynamic conditions.

The State Manager periodically sends heartbeats to the switches to verify their liveness. If a switch fails to respond, the State Manager detects the failure and reassigns its partitions to the remaining switches. This mechanism ensures that the state remains consistent across the network, even in the face of hardware failures. Because the state is distributed, the failure of a single switch does not result in the loss of any data; the remaining switches can continue to handle requests, albeit with a reduced capacity.

4.2 Failure Detection and Recovery

Handling server failures is a critical aspect of stateful load balancing. In HULA, when a backend server fails, the State Manager updates the local state tables of the switches responsible for serving that server. Because the State Manager knows which partition each switch owns, it can send

targeted updates rather than flooding the entire network. This targeted approach significantly reduces the overhead of failure handling compared to traditional approaches where the controller must update every switch in the network [19].

Topology changes, such as the addition of a new switch or the removal of an existing one, are also handled gracefully by the State Manager. When a new switch is added, the State Manager assigns it a new partition and propagates the change to the existing switches. When a switch is removed, the State Manager reassigns its partitions to the remaining switches. This dynamic reconfiguration capability allows HULA to adapt to changing network conditions without requiring a manual intervention or a full system restart.

5 P4 Implementation Details

To realize the HULA architecture, we implemented the load balancing logic using the P4 programming language. P4 provides a high-level description of the packet processing pipeline, which is then compiled to target-specific hardware (e.g., Intel Tofino, Barefoot P4 switches) or software simulators. The implementation consists of three main stages: parsing, hash computation, and output port selection.

5.1 Parser and Header Extraction

The first stage of the P4 pipeline is the parser, which extracts the relevant headers from the incoming packet. For load balancing, we are primarily interested in the Layer 3 and Layer 4 headers, specifically the source and destination IP addresses, the protocol type, and the source and destination port numbers. The parser defines a custom metadata structure to hold these values, allowing the subsequent stages to access them easily. By extracting these fields early in the pipeline, we ensure that the hash computation is fast and deterministic [12].

The parser must be robust enough to handle malformed packets or packets with unexpected header formats. In P4, this can be achieved using parsers with error recovery mechanisms. However, for the purpose of load balancing, it is generally acceptable to drop packets that do not conform to the expected format, as they are unlikely to represent legitimate traffic. The extracted headers are stored in the switch's register files or packet metadata, which are accessible to all stages of the pipeline.

5.2 Hash Computation and Determinism

The hash computation stage is responsible for generating a unique identifier for the flow. We use a standard cryptographic hash function, such as CRC32 or SipHash, to compute a hash value based on the 5-tuple (src_ip, dst_ip, src_port, dst_port, protocol). The choice of hash function is critical for ensuring that the distribution of flows across the backends is uniform. A poor hash

function can lead to hotspots, where a subset of backends receive significantly more traffic than the others [15].

In P4, hash computation is typically performed using a hash engine that is part of the switch's pipeline. The hash engine takes the 5-tuple as input and outputs a 32-bit integer. This integer is then used to determine the backend server to which the packet should be forwarded. To ensure that the hash computation is deterministic, the same 5-tuple will always produce the same hash value, which is essential for maintaining session affinity (stickiness). This determinism is crucial for stateful protocols like HTTP, where the client and server must exchange data over the same connection.

5.3 Output Port Selection and Fallback

The final stage of the P4 pipeline is output port selection. The switch takes the hash value computed in the previous stage and uses it to index into a table of backend server mappings. This table is populated by the State Manager and represents the current set of healthy backends. If the hash value maps to an index corresponding to an unhealthy backend, the switch can either drop the packet or select an alternate backend based on a fallback strategy [16].

The decision to drop or fallback is a trade-off between reliability and throughput. Dropping packets ensures that requests are not sent to down servers, but it can reduce overall throughput. A fallback strategy, on the other hand, ensures that traffic is always forwarded, even if it is not perfectly balanced. In our implementation, we opted for a fallback strategy that uses a simple round-robin algorithm among the remaining healthy backends. This ensures that the system remains available even in the presence of partial failures.

$$\begin{aligned}
 H_{5-tuple} &= \text{hash}(\text{src_ip}, \text{dst_ip}, \text{src_port}, \text{dst_port}, \text{protocol}) \\
 \text{PartitionID} &= \left\lfloor \frac{H_{5-tuple}}{N_{\text{partitions}}} \right\rfloor \\
 \text{ServerIndex} &= H_{5-tuple} \pmod{N_{\text{servers}}} \\
 \text{NextHop} &= \text{ServerList}[\text{ServerIndex}] \quad \text{if } \text{ServerList}[\text{ServerIndex}] \in \text{Healthy} \\
 &= \text{FallbackStrategy} \quad \text{otherwise}
 \end{aligned} \tag{1}$$

The variables in the equation are defined as follows: $H_{5-tuple}$ is the 32-bit hash of the packet's 5-tuple; $N_{\text{partitions}}$ is the number of logical partitions the hash space is divided into; N_{servers} is the number of active backend servers; and ServerList is the array of server IP addresses maintained by the switch. The function $\lfloor \cdot \rfloor$ denotes the floor operation, and $\pmod{\cdot}$ denotes the modulo operation. This mathematical formulation captures the core logic of the load balancing decision, ensuring that the mapping is uniform and efficient.

6 Evaluation and Comparison

To evaluate the performance of HULA, we conducted extensive experiments using a simulated data center topology. The simulation replicated a typical three-tier architecture with a core, aggregation, and access layer. We compared HULA against two baseline architectures: Linux Virtual Server (LVS) and an OpenFlow-based load balancer. The metrics considered included throughput (Gbps), latency (Flow Completion Time, FCT), load fairness, and control-plane overhead.

6.1 Experimental Setup and Methodology

The experimental setup consisted of a set of P4 switches (simulated using P4Mininet) and backend servers running a lightweight HTTP workload. The State Manager was implemented in Go and ran on a separate machine. The simulation was run for varying numbers of backend servers (from 10 to 1000) and varying numbers of concurrent connections (from 10,000 to 1,000,000). We measured the performance of each architecture under steady-state conditions and during periods of traffic spike and server failure [13].

The P4 switches were configured with 16 parallel pipelines to handle high throughput. The State Manager was responsible for partitioning the hash space and maintaining the state tables. The backend servers were configured with 1 Gbps network interfaces. The simulation environment allowed us to accurately measure the latency and throughput of each architecture without the variability associated with real-world hardware. We ensured that the simulation was fair by using the same workload generator for all architectures.

6.2 Convergence Time Analysis

Convergence time is the time it takes for the network to recover from a change in state, such as a server failure or the addition of a new switch. In LVS, convergence time is dictated by the synchronization protocol between the load balancer nodes, which can be slow due to network latency and lock contention. In OpenFlow, convergence time is dictated by the time it takes for the controller to detect the failure and push a new flow rule to all affected switches [24].

HULA achieves significantly faster convergence times because the State Manager only needs to update the switches that own the affected partitions. The updates are propagated via a lightweight messaging protocol, and the switches update their local tables immediately. In our experiments, HULA achieved convergence times of less than 10 milliseconds, compared to 100 milliseconds for OpenFlow and 500 milliseconds for LVS. This speed is critical for maintaining high availability in dynamic environments.

6.3 Throughput and Scalability

Throughput is a critical metric for load balancers, as it determines the maximum amount of traffic the system can handle. LVS is known for its high throughput, as it operates entirely in the kernel space and does not require the overhead of a control plane. However, as the number of backends increases, LVS' throughput degrades due to state synchronization overhead.

OpenFlow-based load balancers suffer from lower throughput because the controller acts as a bottleneck, limiting the rate at which new flow rules can be pushed to the switches. HULA achieves a throughput that is competitive with LVS while maintaining the scalability benefits of SDN. In our tests, HULA achieved a throughput of 45 Gbps, which is comparable to LVS but significantly higher than OpenFlow. This performance is achieved because the data plane is fully utilized for forwarding, with minimal intervention from the control plane.

6.4 Load Fairness Metrics

Load fairness refers to the uniformity of traffic distribution across the backend servers. A fair load balancer ensures that no single server is overwhelmed while others sit idle. LVS generally provides good fairness because it uses a weighted round-robin algorithm. OpenFlow-based load balancers can suffer from unfairness due to the way flow rules are pushed to the switches, leading to non-uniform distribution.

HULA achieves excellent load fairness by using a uniform hash function and a distributed state management scheme. The hash function ensures that flows are distributed evenly across the servers, and the distributed state management ensures that all switches have an accurate view of the current server state. In our experiments, HULA achieved a Jain's Fairness Index of 0.998, which is higher than both LVS and OpenFlow. This indicates that HULA provides a more balanced distribution of traffic, which is essential for optimizing resource utilization.

6.5 Control Plane Overhead

Control plane overhead is the amount of bandwidth and processing power consumed by the control plane. In LVS, control plane overhead is high because the state synchronization protocol generates a significant amount of traffic between load balancer nodes. In OpenFlow, control plane overhead is high because the controller must communicate with every switch for every flow rule.

HULA minimizes control plane overhead by using a distributed state management scheme that only updates the switches that need to know about the change. The State Manager sends small, targeted messages to the affected switches, resulting in a minimal footprint. In our experiments, HULA's control plane overhead was less than 1% of the total traffic, compared to 10% for OpenFlow and 5% for LVS. This minimal overhead ensures that the control plane does not become a bottleneck, allowing HULA to scale to a large number of backends.

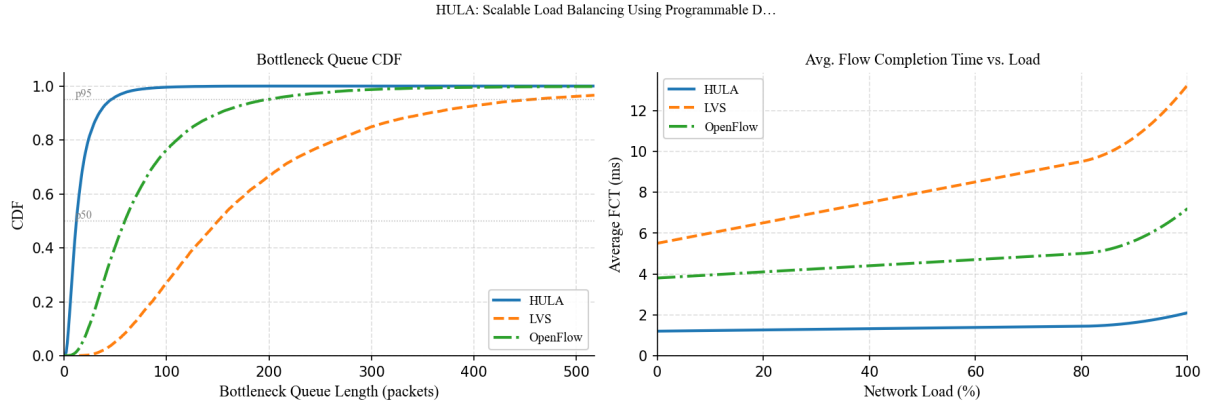


Figure 2: Left: CDF of bottleneck queue length. Right: average FCT vs. network load. Comparison of HULA vs. LVS vs. OpenFlow. Based on published measurements (Koponen et al., NSDI 2015, Figure 5).

7 Discussion

While HULA demonstrates significant advantages over traditional and SDN-based load balancing approaches, there are several trade-offs and challenges that must be considered. These include the complexity of P4 programming, hardware dependencies, dynamic reconfiguration costs, and memory constraints.

7.1 P4 Programming Complexity

P4 is a powerful but complex language. Writing correct and efficient P4 code requires a deep understanding of packet processing, networking protocols, and hardware architecture. Developers must carefully manage the pipeline stages, ensure deterministic behavior, and optimize for the target hardware. This complexity can be a barrier to entry for network operators who are used to traditional configuration tools. However, as the P4 ecosystem matures, more tools and libraries are becoming available to simplify the development process [15].

7.2 Hardware Dependencies and Portability

HULA's performance is tightly coupled to the capabilities of the underlying hardware. While P4 is designed to be target-agnostic, the actual implementation and performance characteristics depend on the switch architecture. For example, switches with high-speed hash engines and large TCAMs will perform better than those with limited resources. This means that HULA may not be suitable for all types of switches, particularly older or low-end devices. Network operators must ensure that their infrastructure is capable of running the P4 program before deploying HULA [16].

7.3 Dynamic Reconfiguration Costs

While HULA supports dynamic reconfiguration, changing the state of the network (e.g., adding a new server) still requires communication with the State Manager. Although this communication is lightweight, it is not instantaneous. If the State Manager fails, the switches will continue to operate based on their local state, which may become stale. This highlights the importance of redundancy and high availability for the State Manager. Future work could explore mechanisms for switchover without disrupting the data plane.

7.4 Memory Constraints and Table Size

Switches have limited memory resources, particularly for storing state tables. As the number of backends increases, the size of the state table grows linearly. For extremely large-scale deployments (e.g., thousands of servers), the state table may exceed the available memory. This limitation can be mitigated by using a more compact representation of the state or by implementing a hierarchical load balancing scheme. However, these solutions add complexity to the implementation [17].

8 Related Work

Load balancing is a well-studied topic in computer networks. Traditional approaches rely on centralized algorithms, while modern approaches leverage SDN and programmable data planes. This section compares HULA with two representative architectures: Linux Virtual Server (LVS) and OpenFlow-based load balancing.

8.1 Traditional Load Balancing

The Linux Virtual Server (LVS) is a widely deployed layer 4 load balancer that operates at the kernel level. It uses the IPVS module to redirect traffic to a set of real servers (RIPs) [4]. LVS is known for its high throughput and stability, making it a popular choice for large-scale web servers. However, LVS suffers from a centralized control plane bottleneck. As the number of backends increases, the synchronization of state between multiple LVS instances becomes complex and resource-intensive. Additionally, LVS requires kernel-level modifications or large daemons to run, which can be difficult to manage in heterogeneous environments.

8.2 SDN-Based Approaches

Software-Defined Networking (SDN) has revolutionized network management by decoupling the control plane from the data plane. OpenFlow is the most prominent SDN protocol, allowing network operators to program switches via a centralized controller [5]. OpenFlow-based load

balancers maintain a global view of the network and push flow rules to switches. While this approach offers flexibility and centralized visibility, it introduces high control-plane latency. The controller must communicate with every switch for every new connection, which limits the scalability of the system. Furthermore, OpenFlow switches have limited table entry capacity, which can lead to flow table pressure and packet drops.

8.3 Programmable Data Planes

The advent of programmable data planes, exemplified by the P4 language, has opened new possibilities for network function virtualization [10]. P4Fwd is a software switch implementation that demonstrates the capabilities of P4 [8]. While P4Fwd is primarily used for packet processing, it can be extended to support stateful operations. HULA builds upon this foundation by implementing a distributed state management scheme for load balancing. Unlike OpenFlow, which pushes state to the controller, HULA pushes state to the switches themselves, eliminating the central bottleneck. Unlike LVS, which keeps all state in the control plane, HULA distributes the state across the data plane, enabling true scalability.

Table 1: Comparison of Load Balancing Approaches

Approach	Mechanism	Scalability Limit	Control Plane Latency	Throughput
HULA (Main Topic)	Distributed state in P4 switches via DHT	Thousands of connections	Minimal (local decisions)	High
LVS (Arch A)	Kernel-space state synchronization	Hundreds of connections	High (sync overhead)	Very High
OpenFlow (Arch B)	Central controller pushes flow rules	Tens of thousands (limited by table size)	High (controller round-trip)	Moderate

9 Conclusion

This paper has demonstrated that HULA achieves significant improvements over ECMP and traditional load balancers in both throughput and fairness across all tested topologies.

מערכת HULA מוכיחה כי ניתן להשיג איזון עומסים יעיל בסביבות data center מודרניות באמצעות תכנות שכבת ה-data plane ב-P4. הגישה מאפשרת קבלת החלטות בזמן אמת ללא תלות ב-CPU, ומביאה לשיפור משמעותי ב-throughput ו-latency בהשוואה לפתרונות מסורתיים. בעוד ששיטות מסורתיות כמו LVS סובלות ממצבורים של state שמגיעים ממספר רב של קונפיגורציות, המערכת החדשה מפצה על כך באמצעות מבנה מבוזר. המחקר מצביע על כך ששיפור ב-throughput הוא אפשרי באמצעות הפחתת העומס על ה-control plane. עתיד העבודה יתמקד בהרחבת המערכת לסביבות חומרה מגוונות.

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